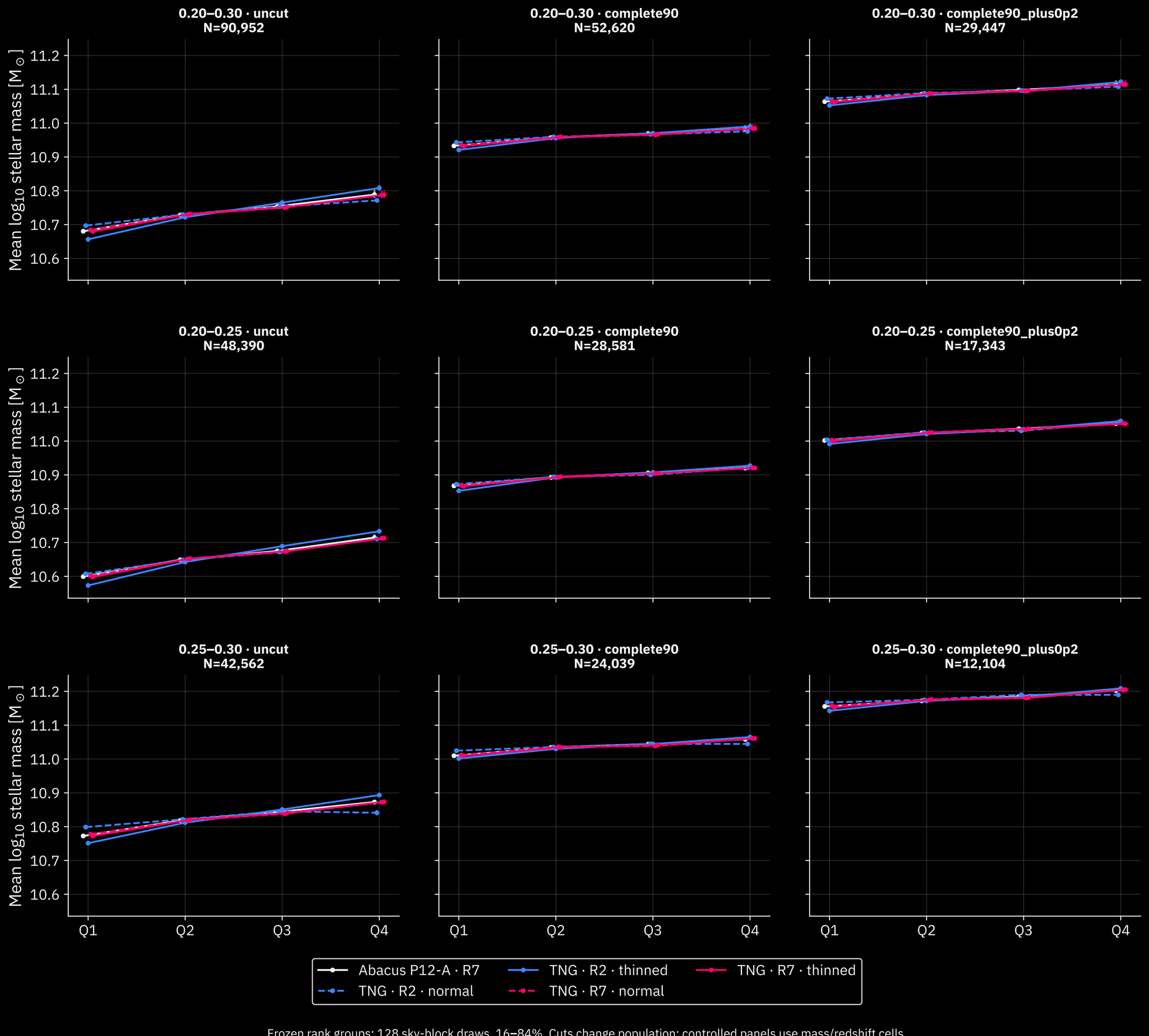
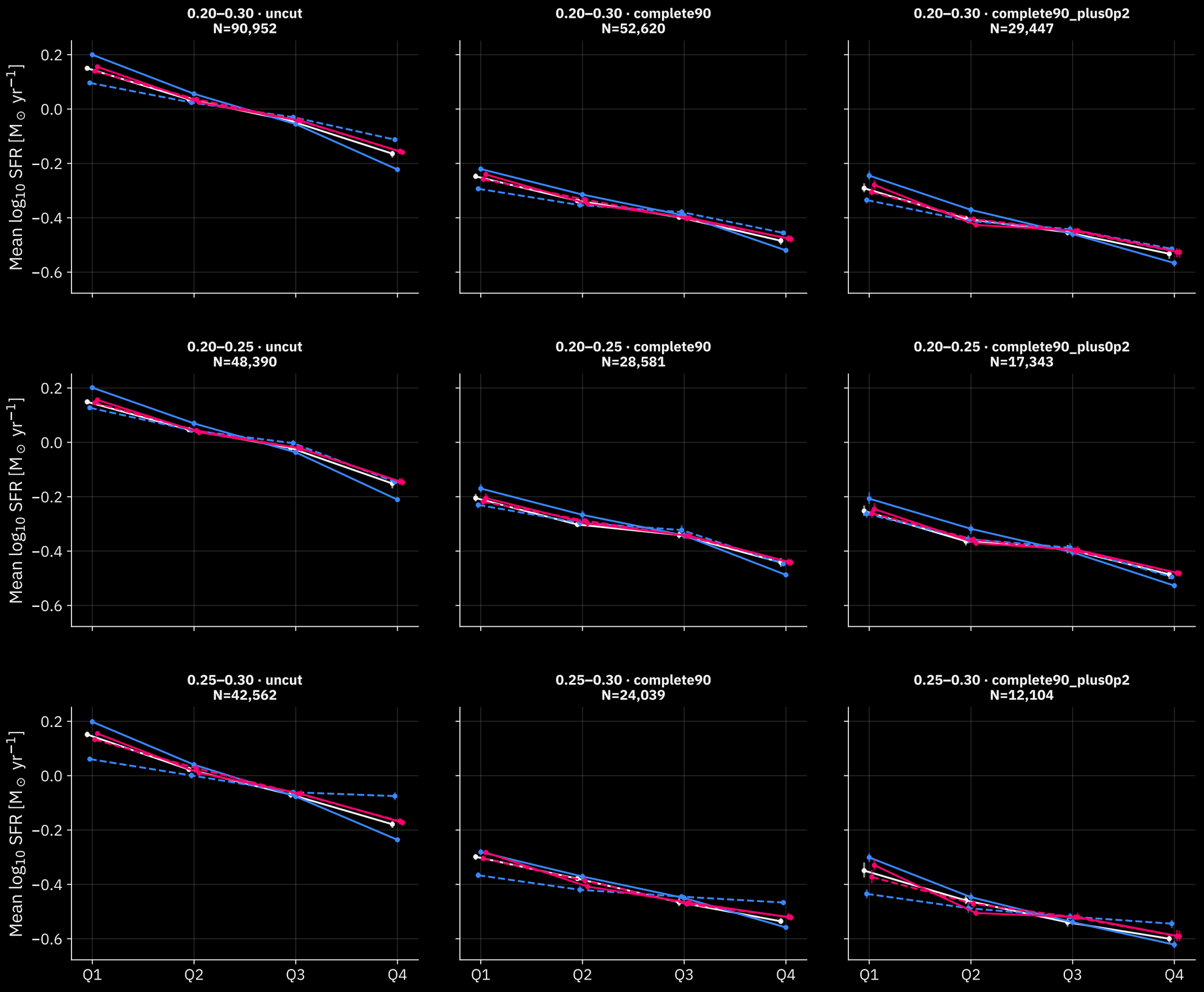
Selection sensitivity: mass · rank_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



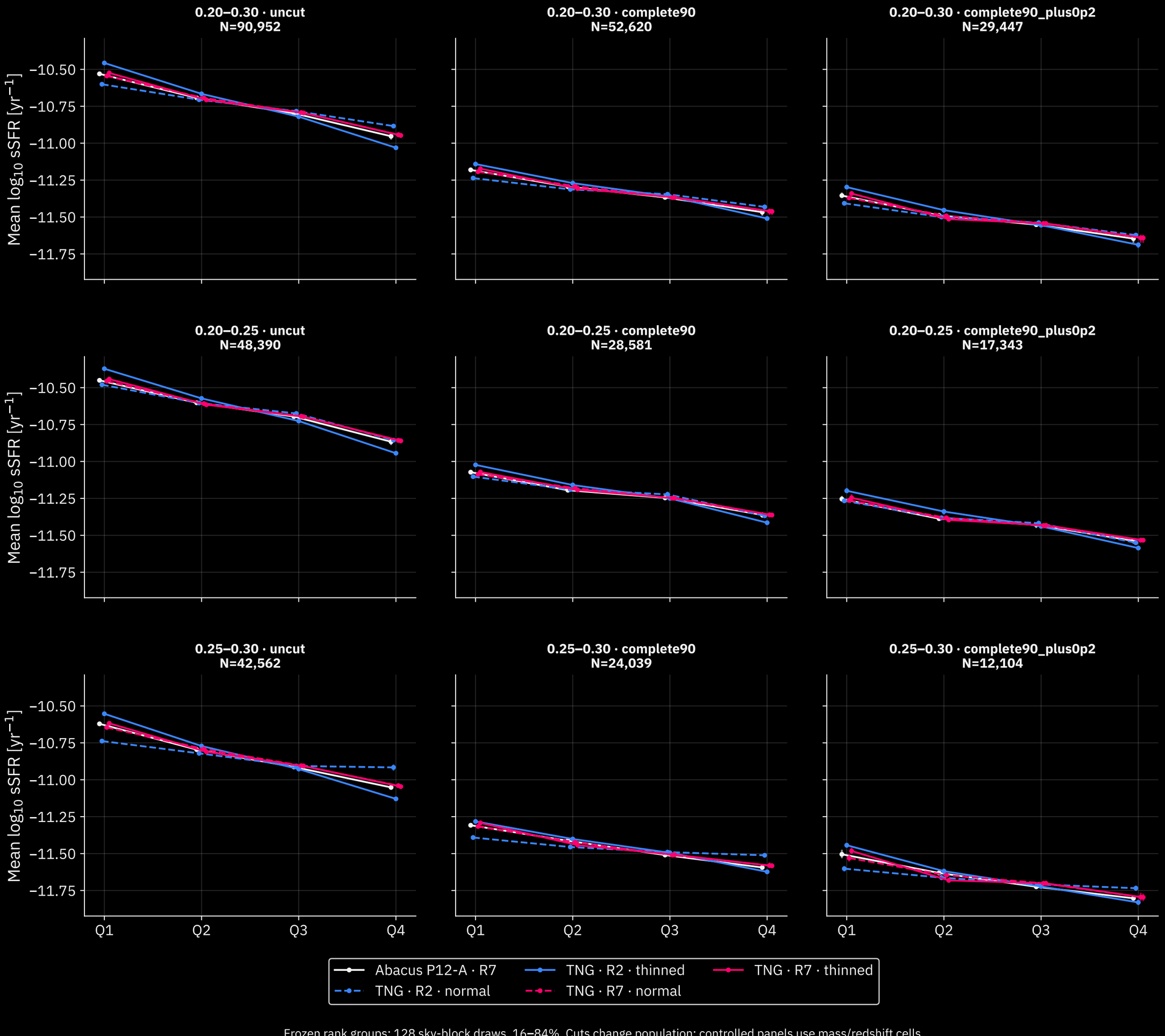
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: sfr · rank_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



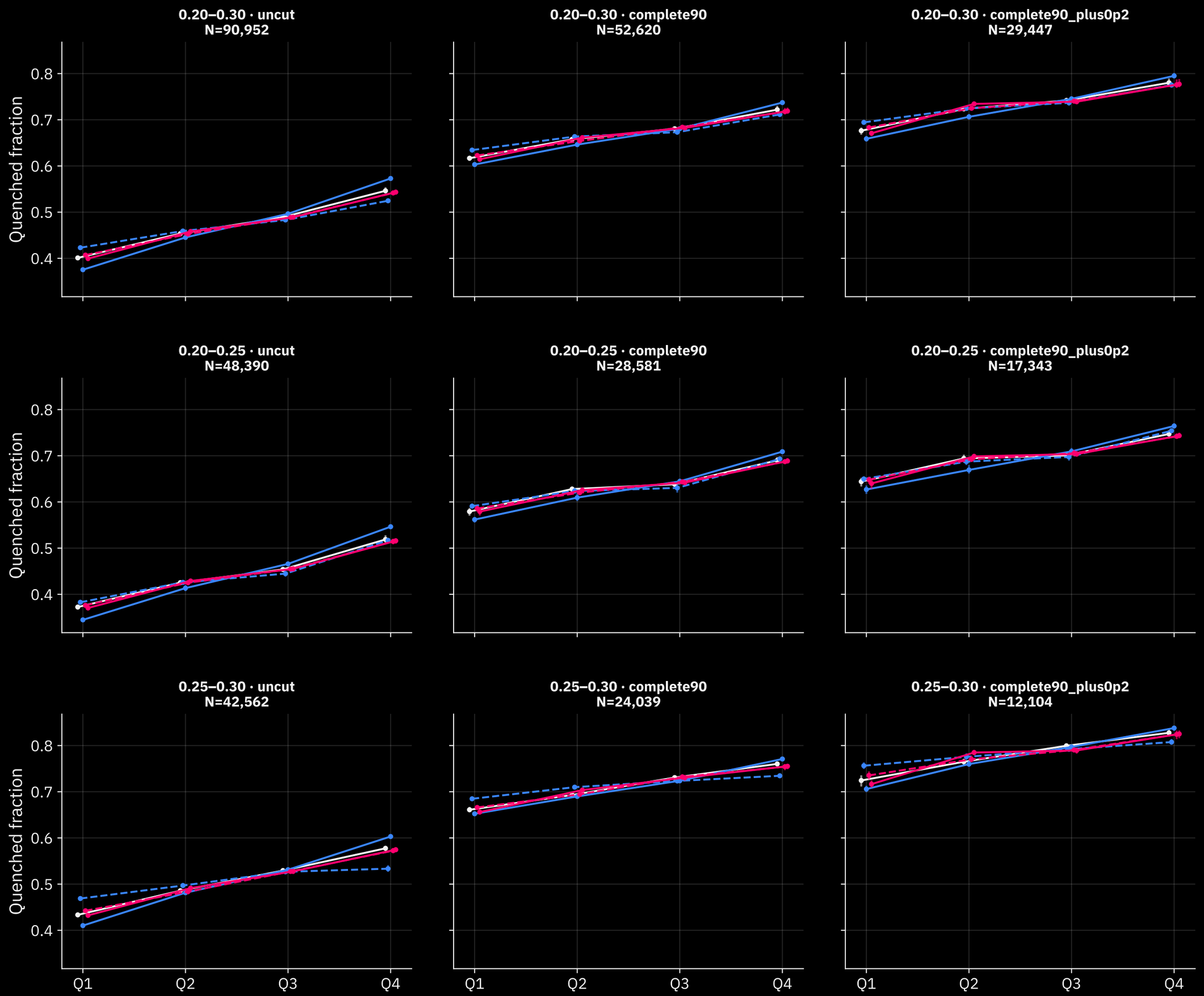
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: ssfr · rank_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



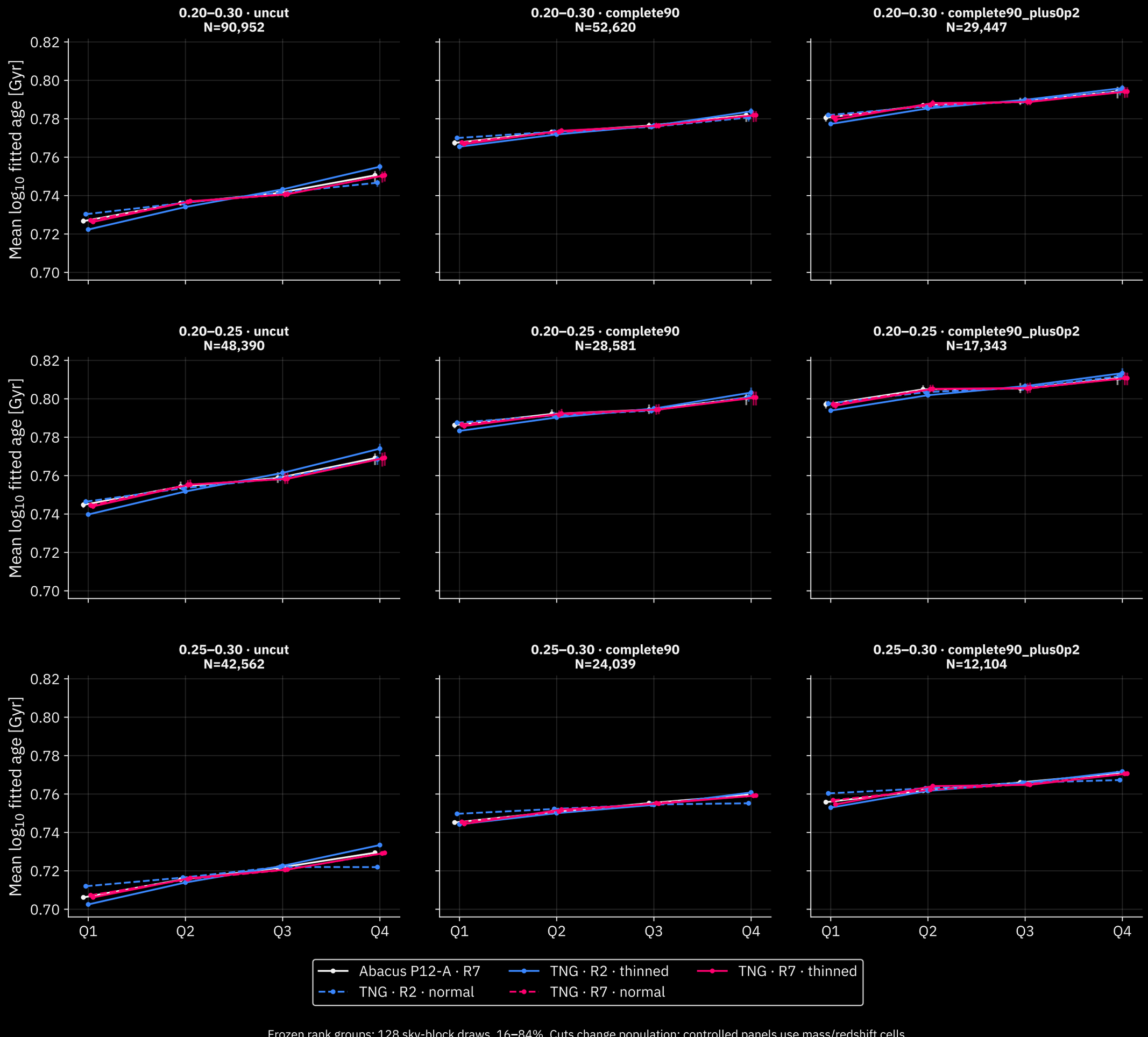
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: quenched · rank_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



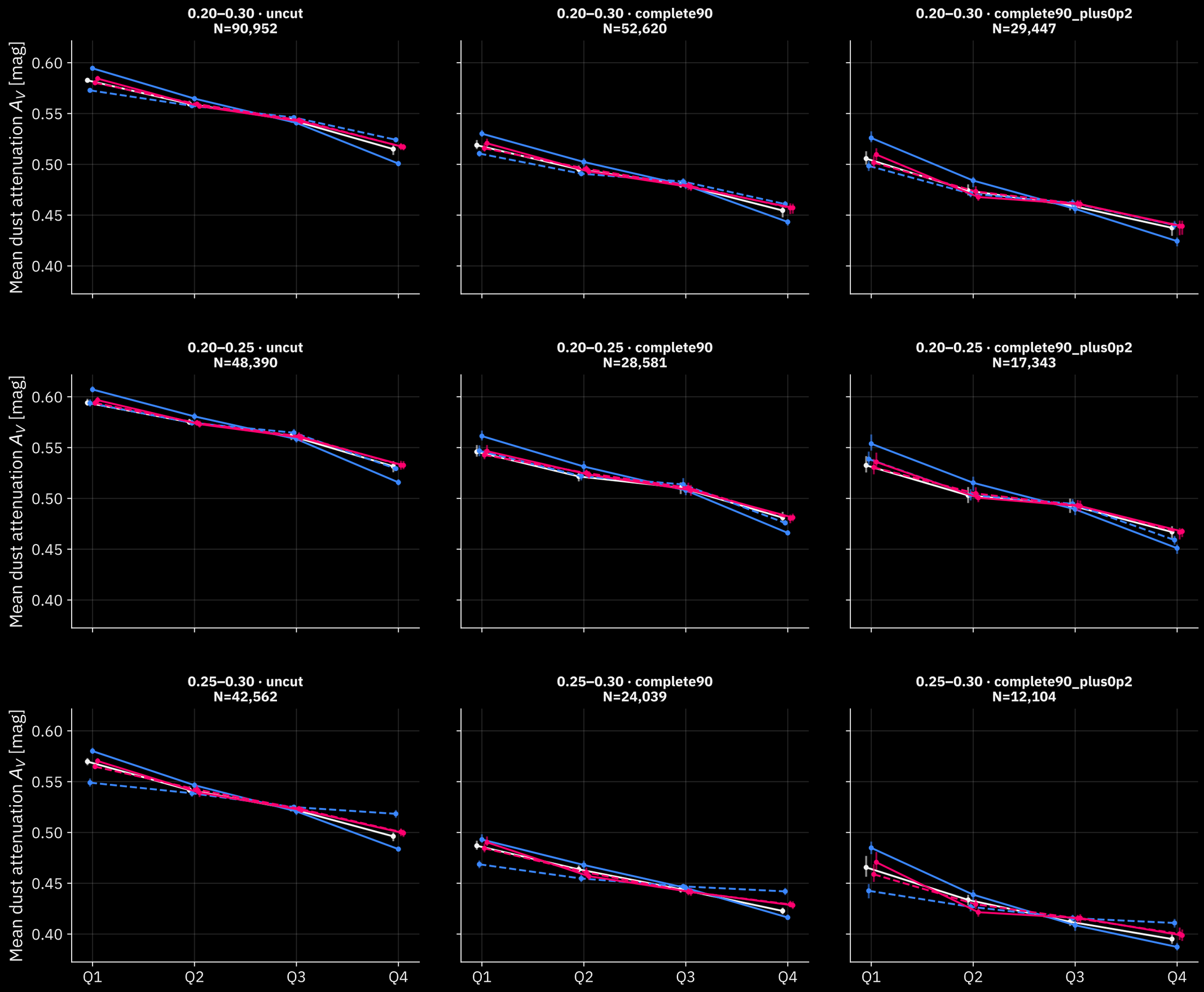
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: age · rank_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



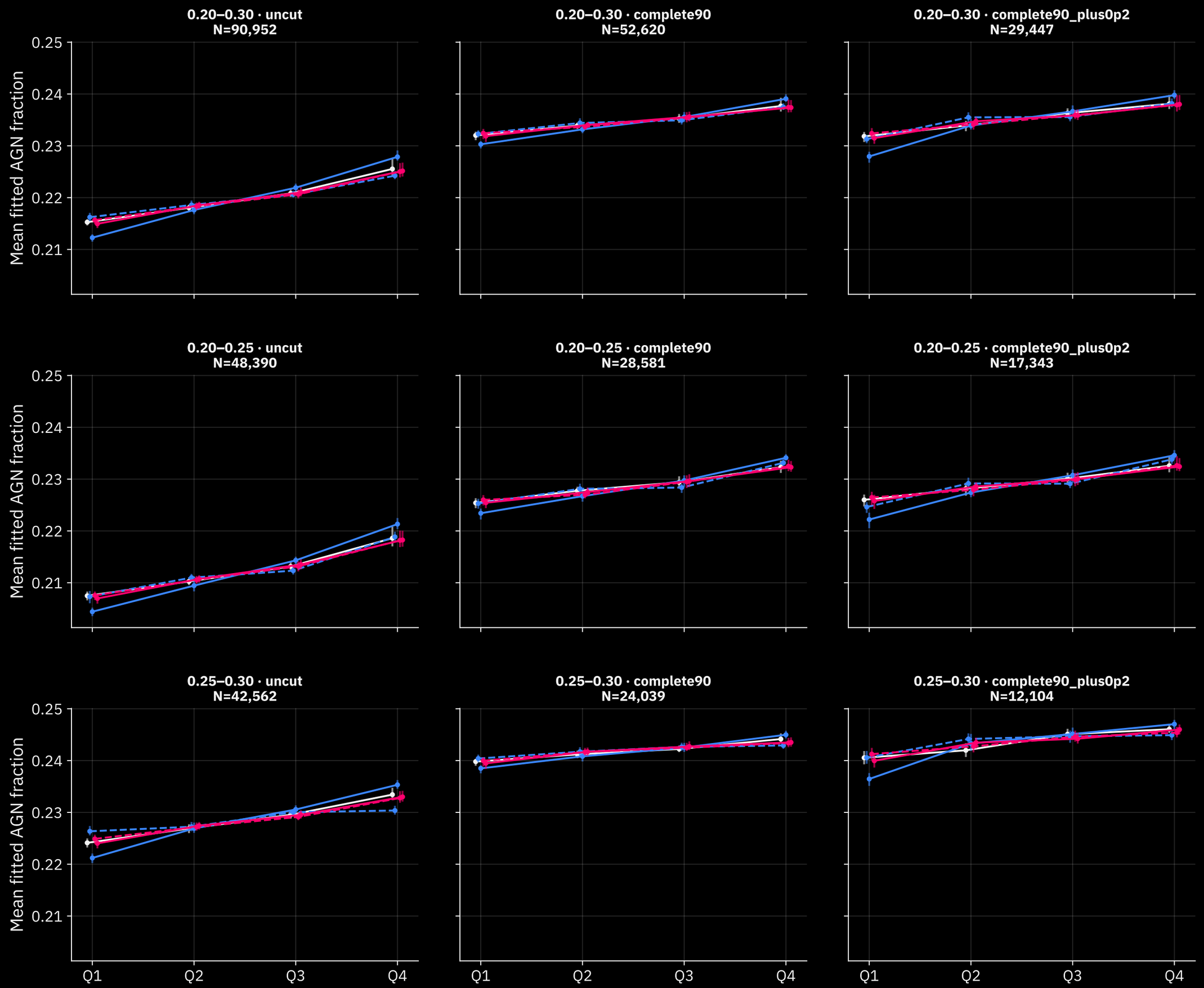
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: av · rank_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



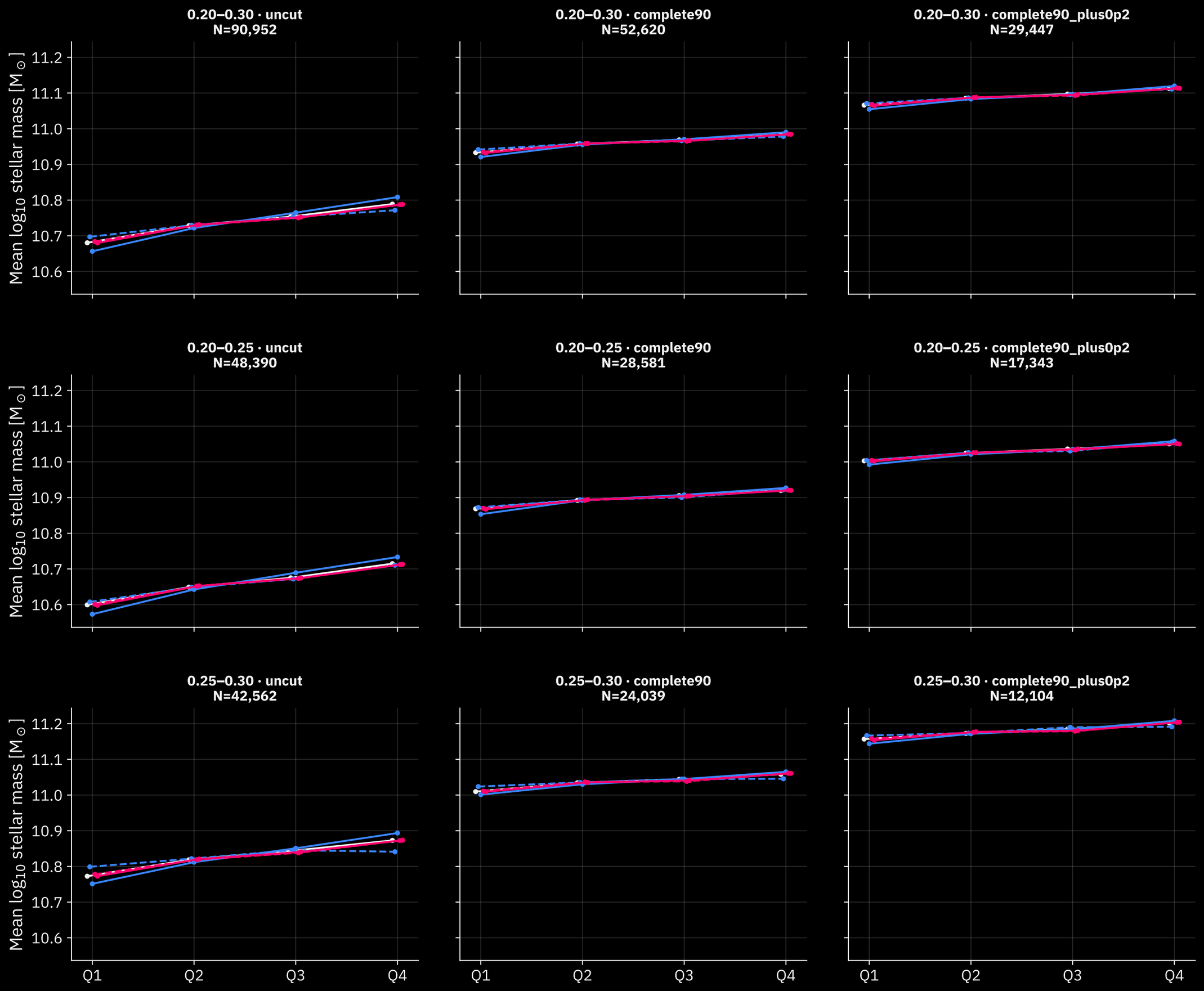
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: fagn · rank_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: mass · rank_controlled
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: sfr · rank_controlled
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel

0.20–0.30 · uncut
N=86,695

0.20–0.30 · complete90
N=50,173

0.20–0.30 · complete90_plus0p2
N=27,003

0.20–0.25 · uncut
N=46,301

0.20–0.25 · complete90
N=27,324

0.20–0.25 · complete90_plus0p2
N=16,089

0.25–0.30 · uncut
N=40,394

0.25–0.30 · complete90
N=22,849

0.25–0.30 · complete90_plus0p2
N=10,914

Mean $\log_{10}$ SFR [$M_\odot$ yr^{-1}]

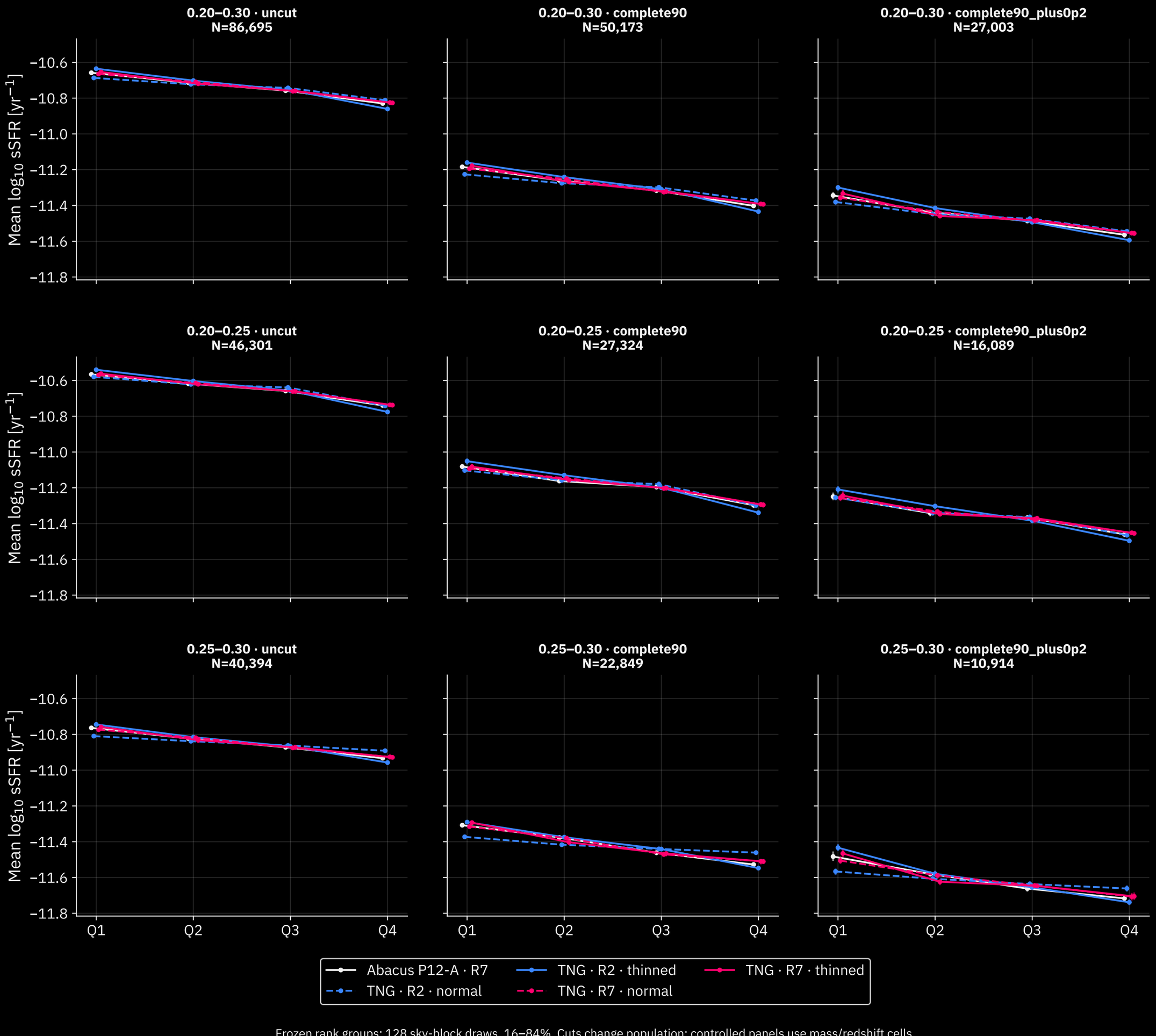
0.0
−0.2
−0.4
−0.6

Q1 Q2 Q3 Q4

Abacus P12-A · R7
TNG · R2 · normal
TNG · R2 · thinned
TNG · R7 · normal
TNG · R7 · thinned

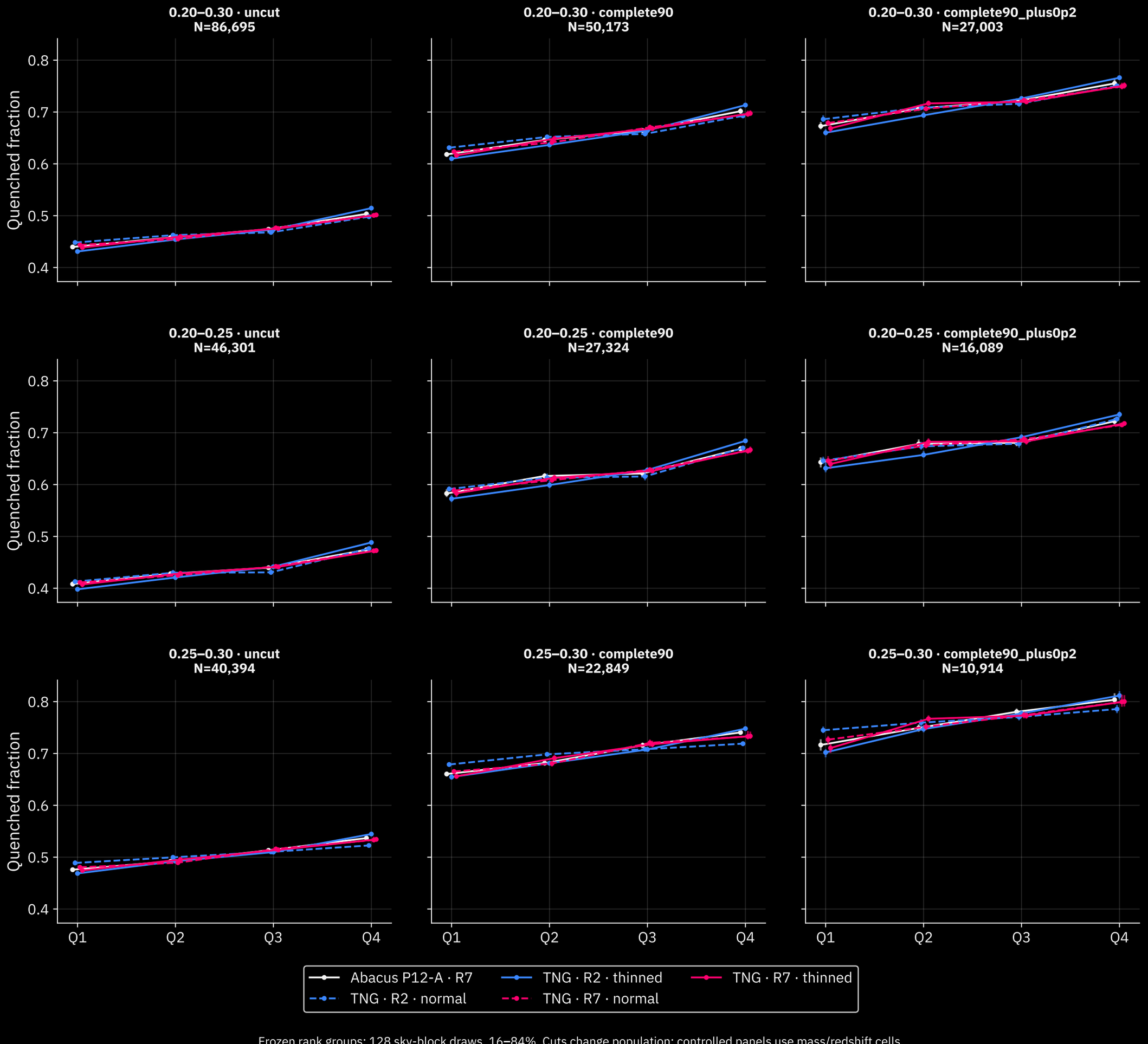
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: ssfr · rank_controlled
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



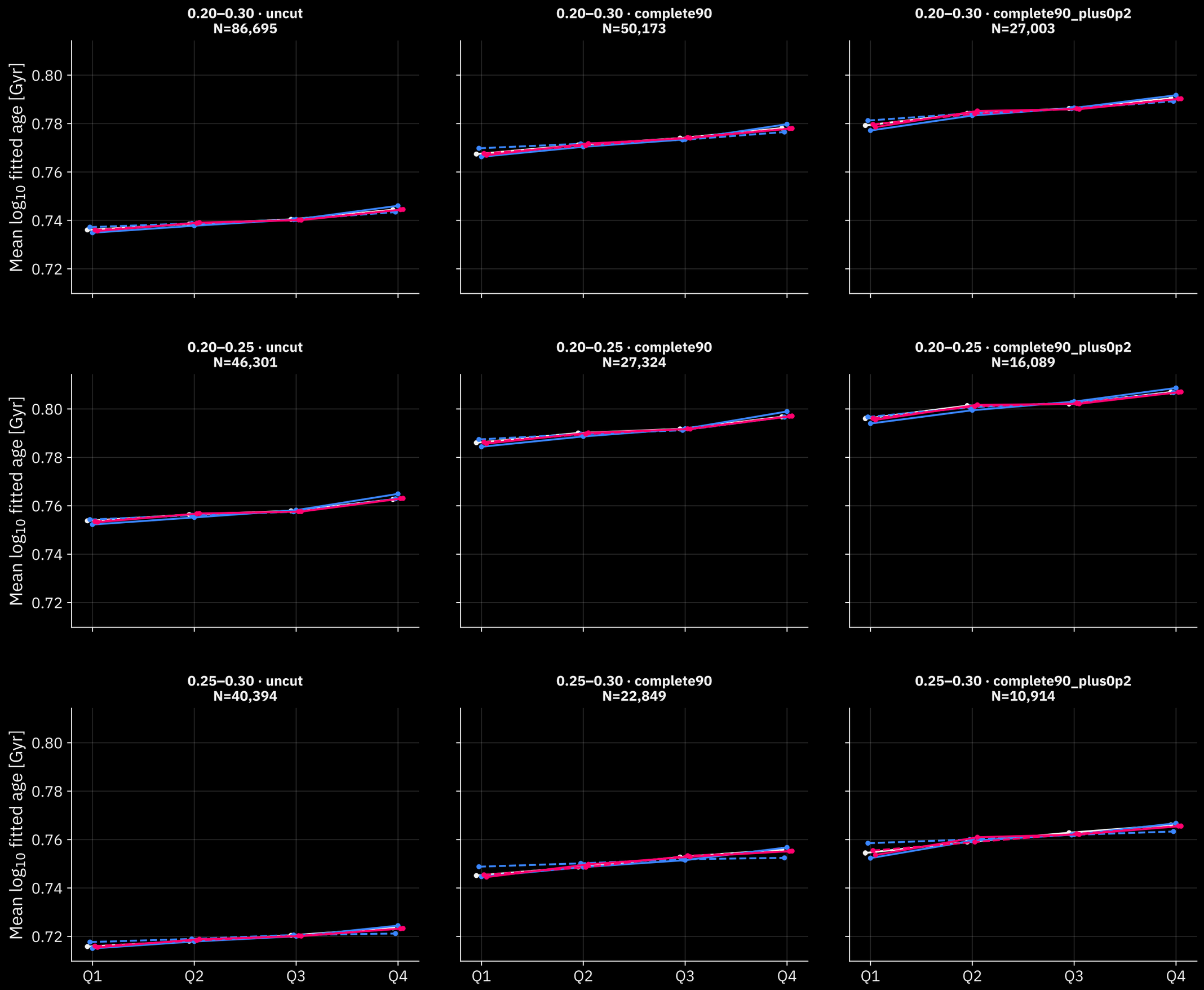
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: quenched · rank_controlled
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



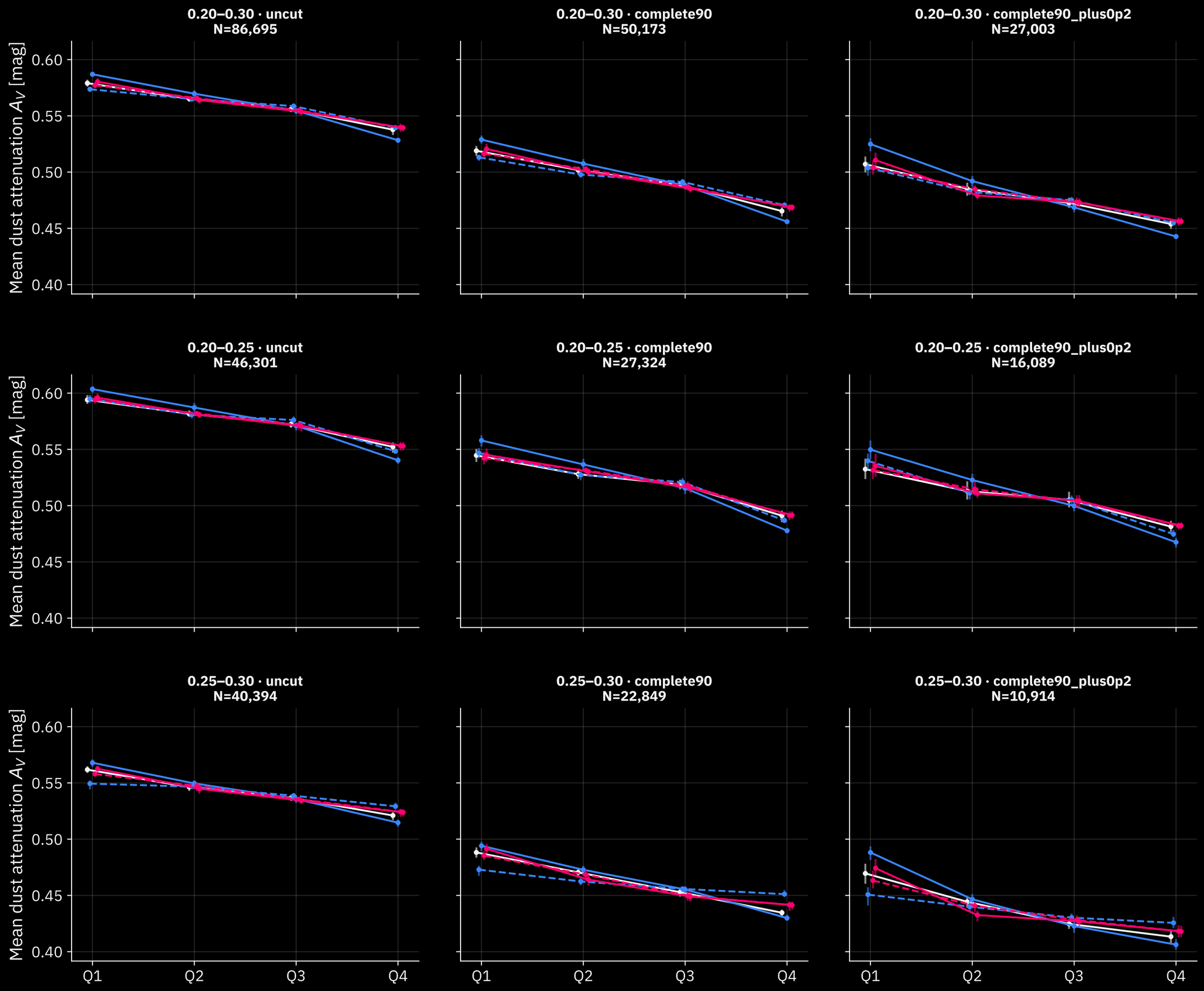
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: age · rank_controlled
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



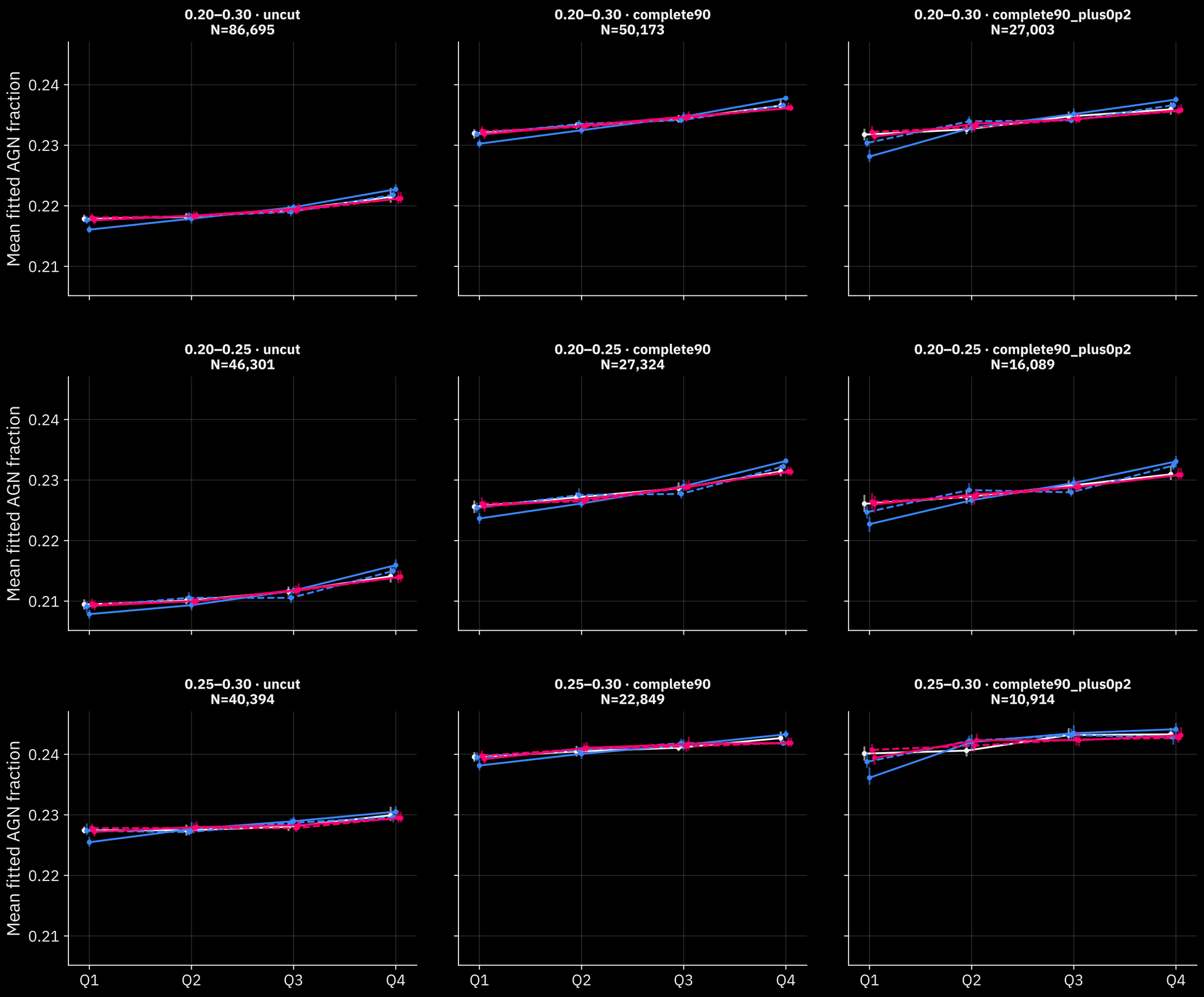
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: av · rank_controlled
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: fagn · rank_controlled
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: mass · class_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel

0.20–0.30 · uncut
N=90,952

0.20–0.30 · complete90
N=52,620

0.20–0.30 · complete90_plus0p2
N=29,447

0.20–0.25 · uncut
N=48,390

0.20–0.25 · complete90
N=28,581

0.20–0.25 · complete90_plus0p2
N=17,343

0.25–0.30 · uncut
N=42,562

0.25–0.30 · complete90
N=24,039

0.25–0.30 · complete90_plus0p2
N=12,104

Mean $\log_{10}$ stellar mass [$M_\odot$]

11.2
11.1
11.0
10.9
10.8
10.7
10.6
10.5

Void
Wall
Filament
Cluster

Abacus P12-A · R7
TNG · R2 · thinned
TNG · R7 · thinned
TNG · R2 · normal
TNG · R7 · normal

Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: sfr · class_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel

0.20–0.30 · uncut
N=90,952

0.20–0.30 · complete90
N=52,620

0.20–0.30 · complete90_plus0p2
N=29,447

0.20–0.25 · uncut
N=48,390

0.20–0.25 · complete90
N=28,581

0.20–0.25 · complete90_plus0p2
N=17,343

0.25–0.30 · uncut
N=42,562

0.25–0.30 · complete90
N=24,039

0.25–0.30 · complete90_plus0p2
N=12,104

Mean $\log_{10}$ SFR [$M_\odot$ yr^{-1}]

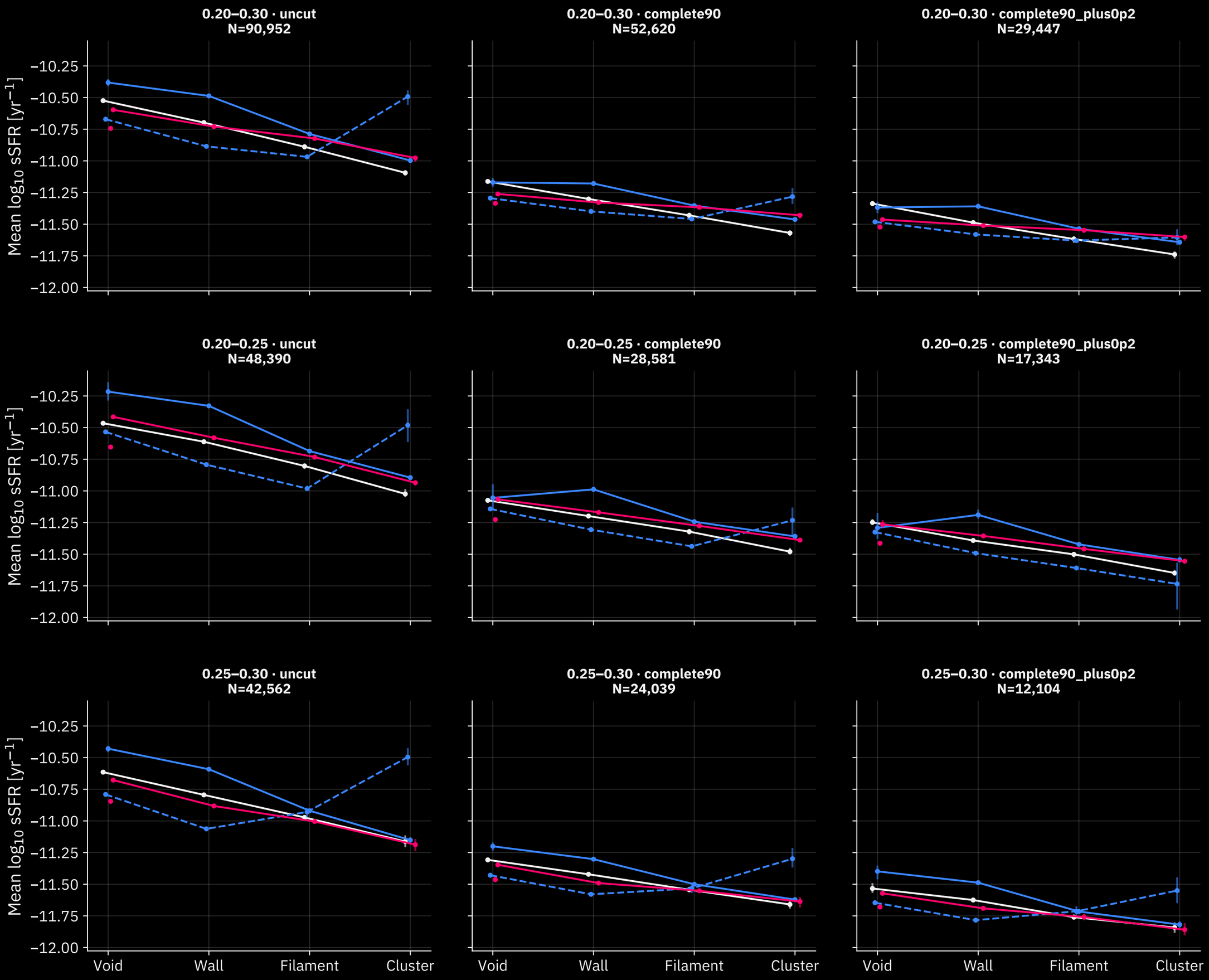
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8

Void
Wall
Filament
Cluster

Abacus P12-A · R7
TNG · R2 · thinned
TNG · R7 · thinned
TNG · R2 · normal
TNG · R7 · normal

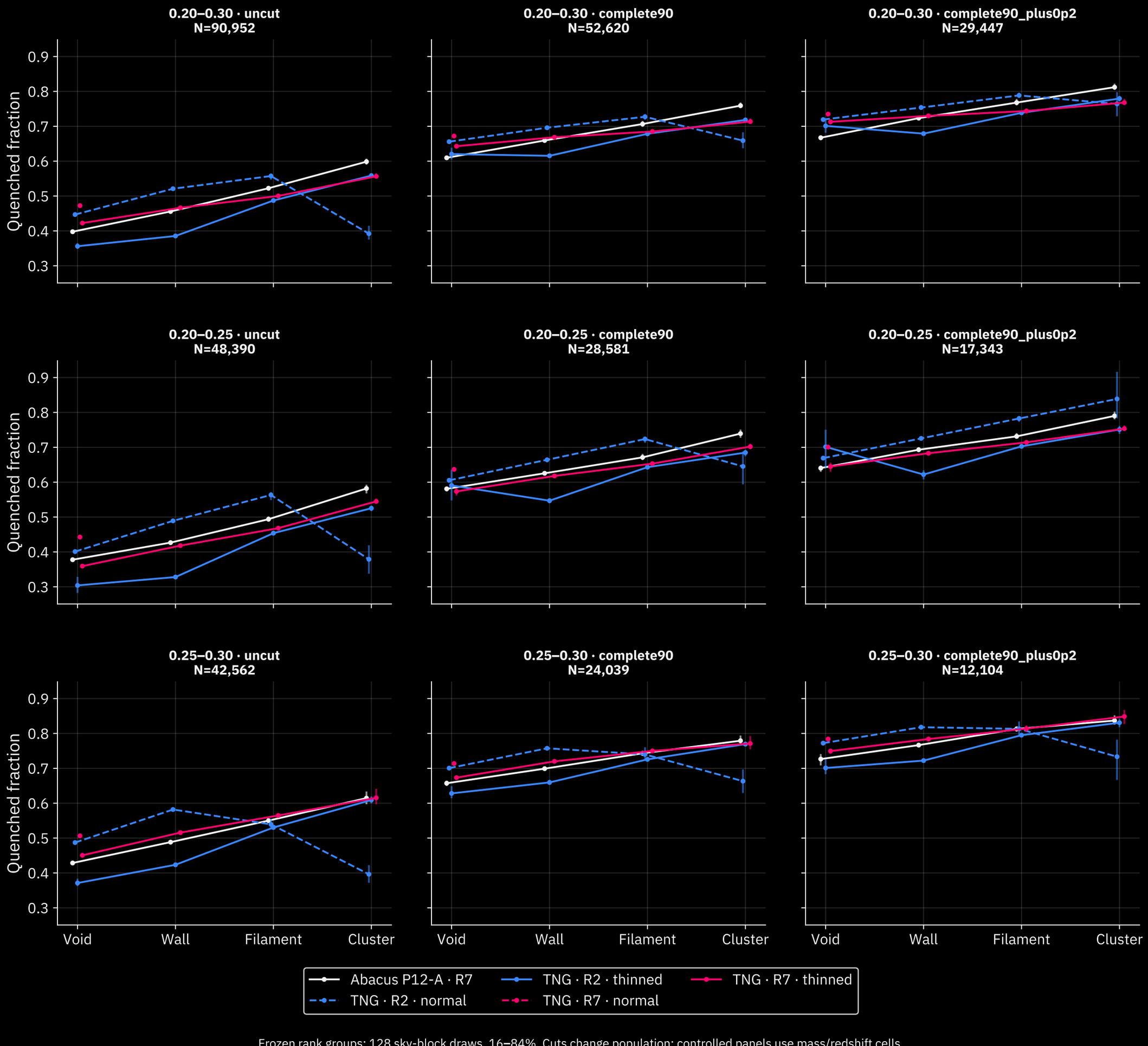
Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: ssfr · class_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: quenched · class_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: age · class_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel

0.20–0.30 · uncut
N=90,952

0.20–0.30 · complete90
N=52,620

0.20–0.30 · complete90_plus0p2
N=29,447

0.20–0.25 · uncut
N=48,390

0.20–0.25 · complete90
N=28,581

0.20–0.25 · complete90_plus0p2
N=17,343

0.25–0.30 · uncut
N=42,562

0.25–0.30 · complete90
N=24,039

0.25–0.30 · complete90_plus0p2
N=12,104

Mean $\log_{10}$ fitted age [Gyr]

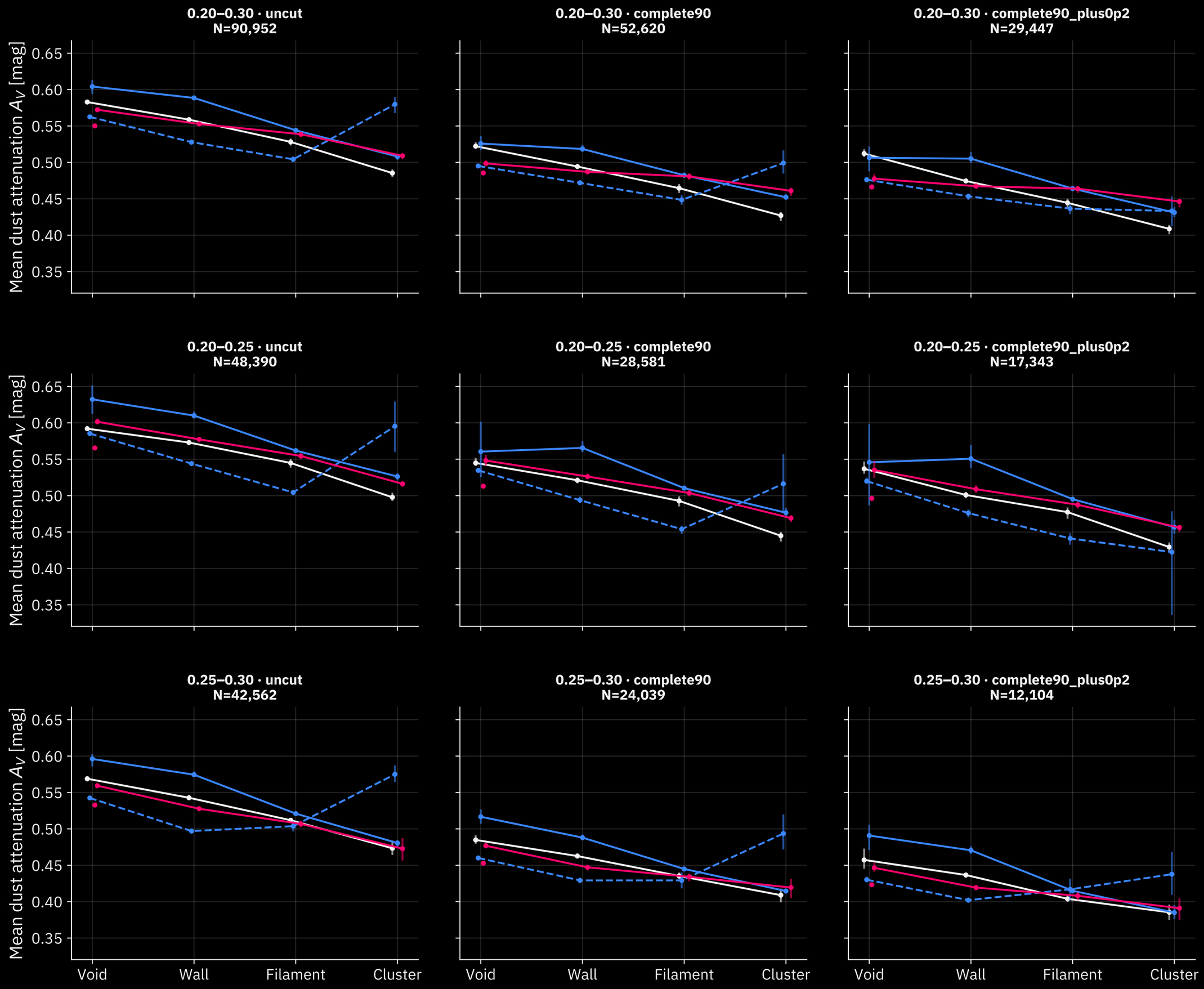
0.82
0.80
0.78
0.76
0.74
0.72
0.70

Void
Wall
Filament
Cluster

Abacus P12-A · R7
TNG · R2 · thinned
TNG · R7 · thinned
TNG · R2 · normal
TNG · R7 · normal

Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: av · class_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel



Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.

Selection sensitivity: fagn · class_raw
TNG local-feature MLPs and Abacus P12-A; same Loa galaxies per panel

0.20–0.30 · uncut
N=90,952

0.20–0.30 · complete90
N=52,620

0.20–0.30 · complete90_plus0p2
N=29,447

0.20–0.25 · uncut
N=48,390

0.20–0.25 · complete90
N=28,581

0.20–0.25 · complete90_plus0p2
N=17,343

0.25–0.30 · uncut
N=42,562

0.25–0.30 · complete90
N=24,039

0.25–0.30 · complete90_plus0p2
N=12,104

Mean fitted AGN fraction

0.26
0.24
0.22
0.20

Void
Wall
Filament
Cluster

Abacus P12-A · R7
TNG · R2 · thinned
TNG · R7 · thinned
TNG · R2 · normal
TNG · R7 · normal

Frozen rank groups; 128 sky-block draws, 16–84%. Cuts change population; controlled panels use mass/redshift cells.